

Make Games, Learn Math

Hyosang (Henry) Kang

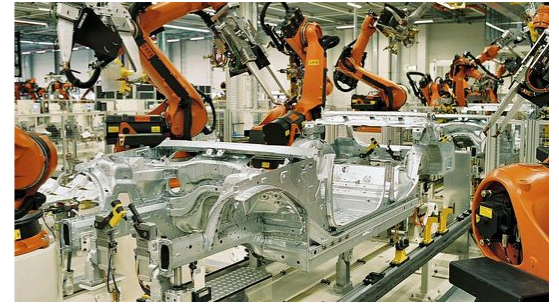
Department of Liberal Arts and Science, DGIST, South Korea

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Department of Infrastructure Engineering, University of Melbourne

Engineering

Create a solution for real-world by simulation.



Game engines

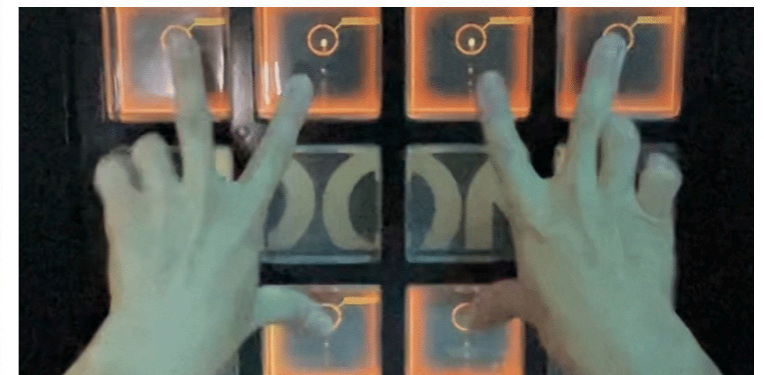
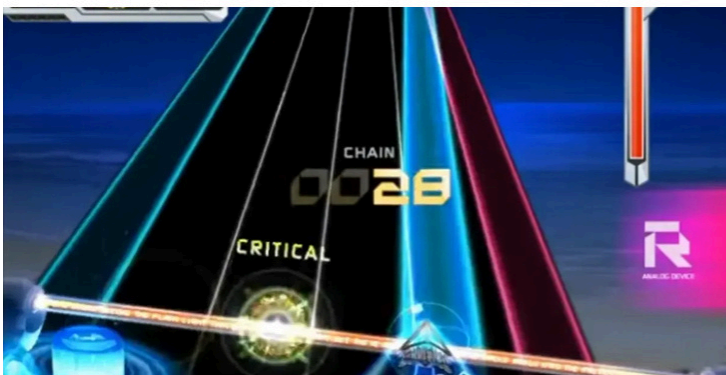
Create a game for players by emulation.



How It Started and Developed

The Undergraduate Group Research Program (UGRP)

- An accurate rhythm-game level algorithm (2019)



A Mega UGRP Project in 2020

- Making games with unforgettable stories

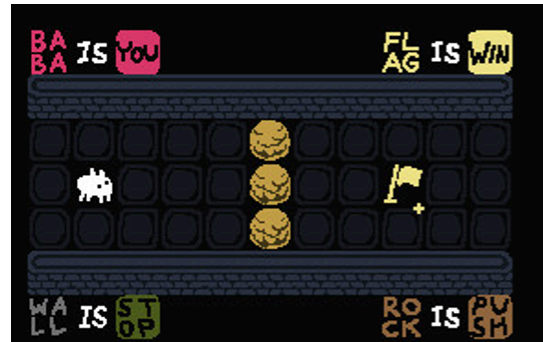
Fun



Story



Challenge



Immersion



Project based learning in math subjects (2021-)

- **2021**(Multivariable Calculus): Coding a math package in **Go**

EM = Engineering Mathematics

- **2022**(EM): Computing, drawing, modeling, estimating with **Go**
- **2023**(EM): Game programming using **Ebiten** with DE
- **2024**(EM): Game development with math ideas (**Ebiten**)
- **2025**(EM): Game developement (**Ebiten**)

Game Making in Math Education

Essentials of Mathematics Education:

1. Mathematical Modeling

- Differential equation: predicting the system
- Functional analysis: Fourier transform, complex analysis, etc

2. Optimization

- Objective function: discrete or continuous
- Algorithm: deterministic, undeterministic, heuristic, etc

Essentials of Game:

1. Goal

- Expectations: **why should I play this game?**
- Objectives: **what should I do in this game?**

2. Rule

- Interface: **how do I play this game?**
- Story: **how can I immerse myself in this game?**

Game Making: Mathematics and Communication

Quantitative decisions

- Scoring, scene transitions, visuals, game logic, game balance
- Mathematical modeling and optimization

Qualitative decisions

- Genre, narrative, visuals (design, illustration)
- Communication and expression

Mathematical Modeling: Differential Equation

- Predicting the systems (over time)
- **Example:** projectile motion

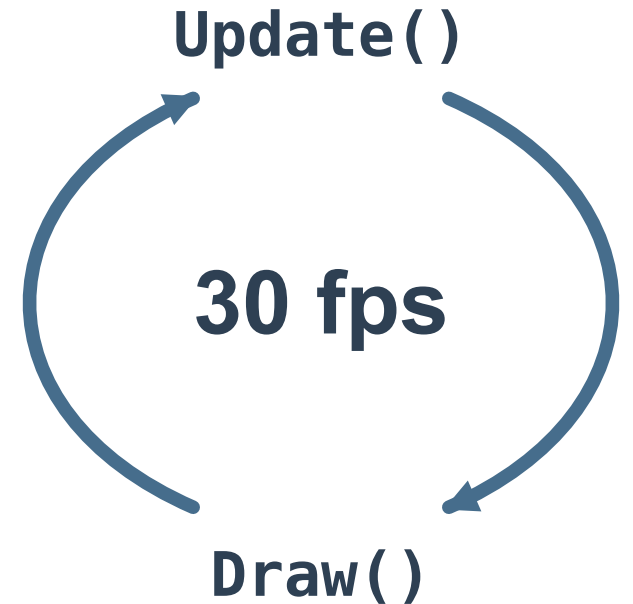
$$\frac{dx}{dt} = v, \quad \frac{d^2y}{dt^2} = -g$$

- Numerical approximation (Euler's method)

$$\Delta x = v\Delta t, \quad \Delta y = \dot{y}\Delta t, \quad \Delta \dot{y} = -g\Delta t$$

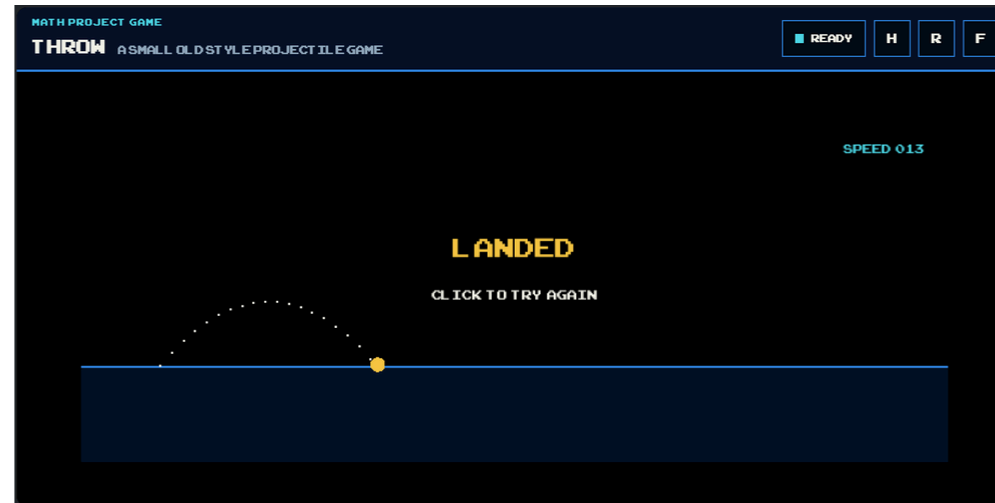
How Game Engine Works (Ebiten)

- **Update()**
 - Take user inputs
 - Update the parameters
- **Draw()**
 - Refresh the screen



A simple game example: Throw!

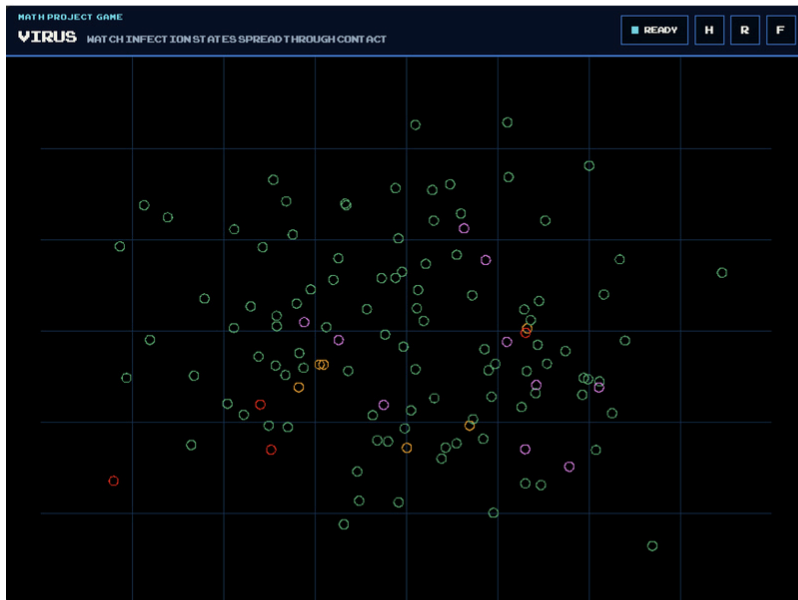
- Interface & Handle scenes → Modeling



Example: Virus!

- Automation (Free agents) → Algorithm

Virus!



League of Legends



StarCraft



Totally Accurate Battle Simulator



Plague Inc.



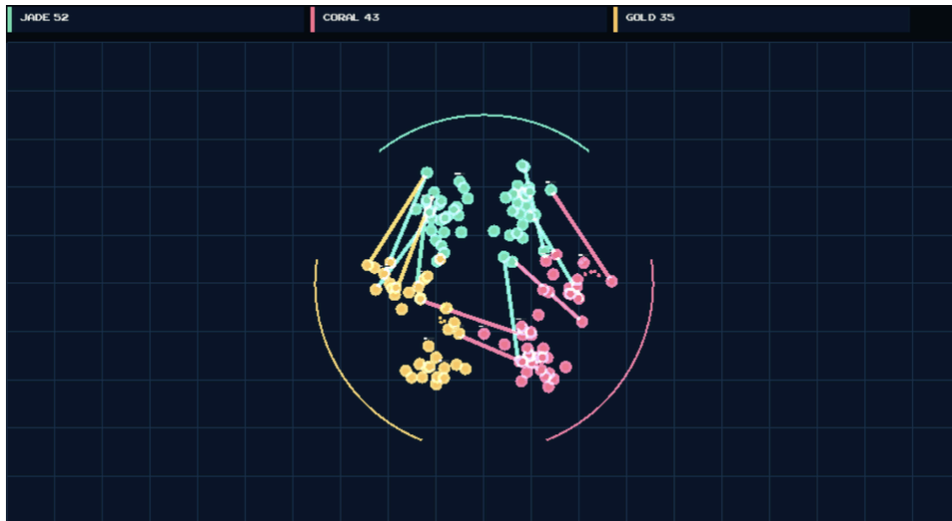
How it Worked

Assessment

- **Organization:** What was the plan of the project and how much was it complete?
- **Mathematics:** What are the underlying mathematical ideas in the project?
- **Creativity:** What make the project stand out?

Engineering Mathematics I: Spring 2022

- **Theme:** Build an agent-based simulation of war or a pandemic and compare the results with numerical solutions.

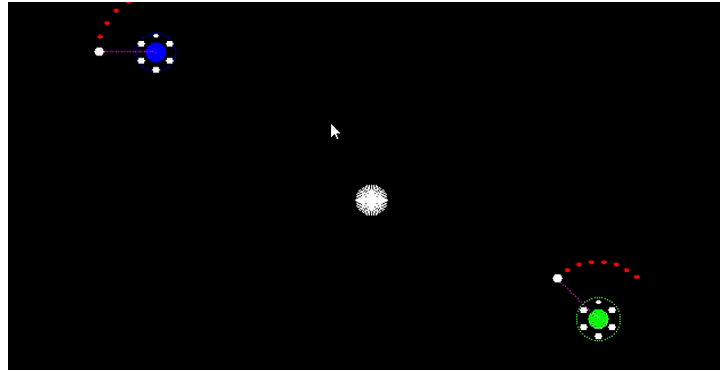


Engineering Mathematics I: Spring 2023

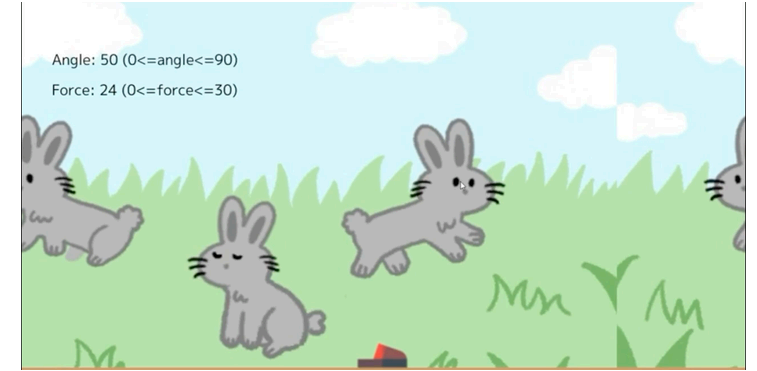
- **Theme:** Develop a game whose outcome can be predicted using ordinary differential equations.



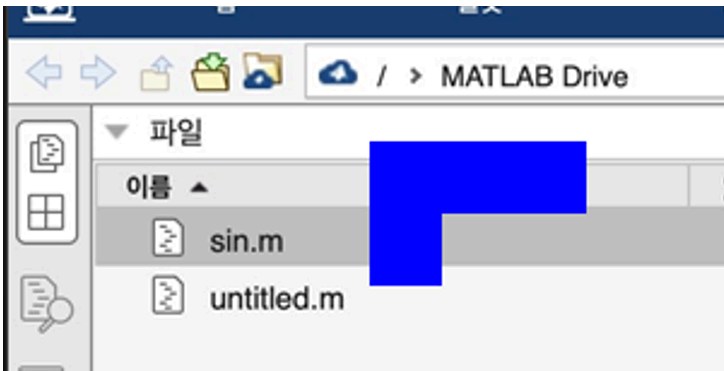
Farm



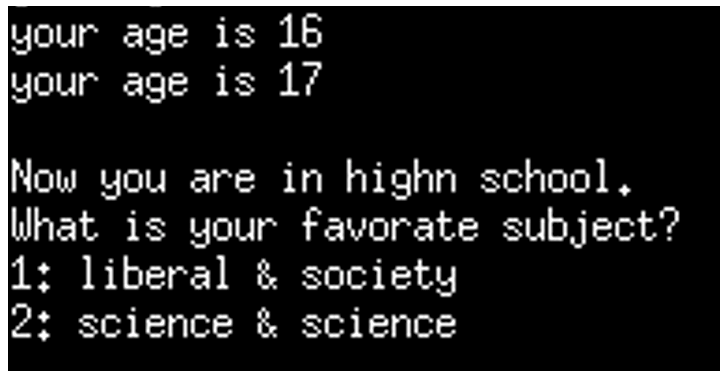
Orbit



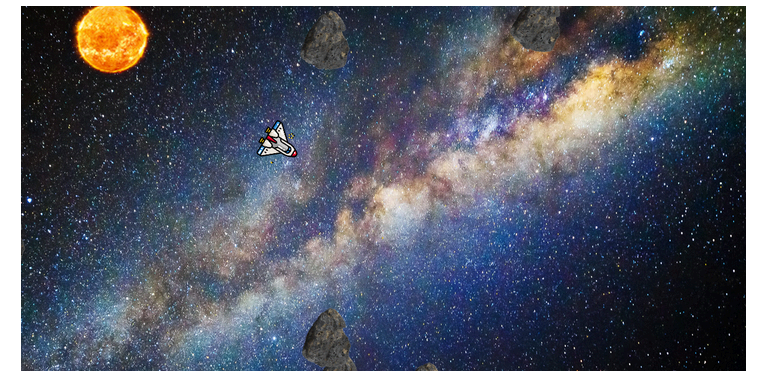
Slipper Toss



Tetris



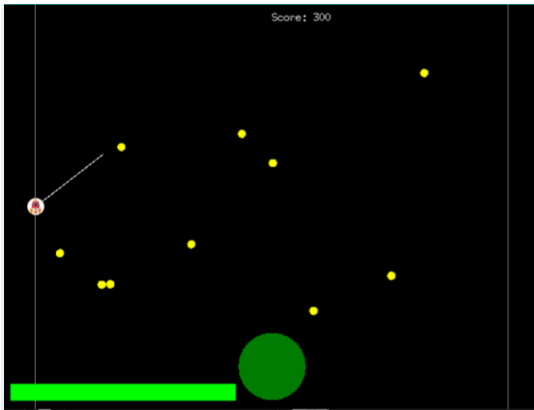
Karma



Rocket Game

Engineering Mathematics I: Spring 2024

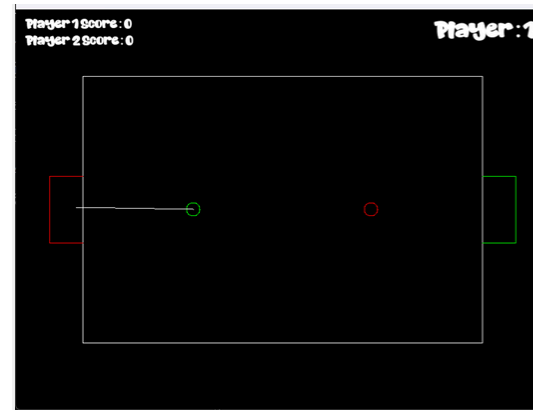
- **Theme:** Develop a game that incorporates any mathematical ideas.



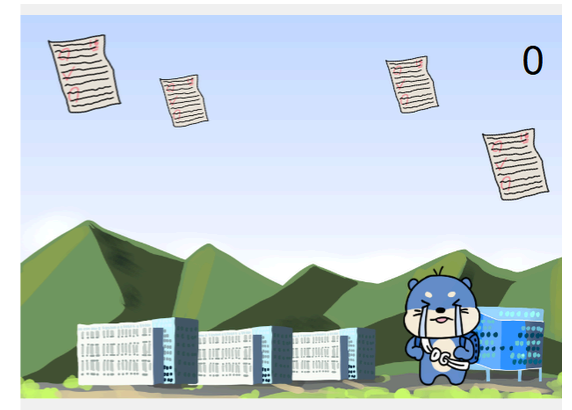
Spaceship game



Are you okay,
professor?

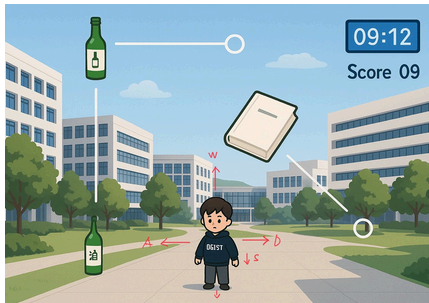
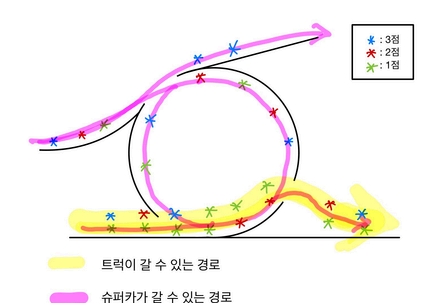
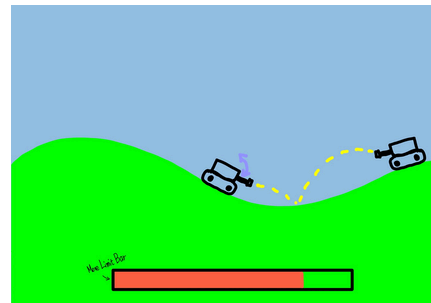
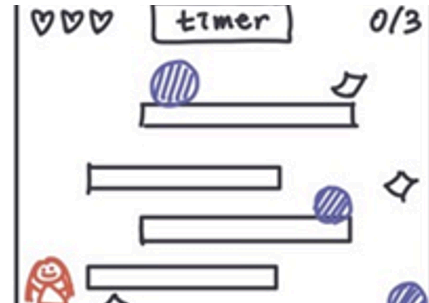
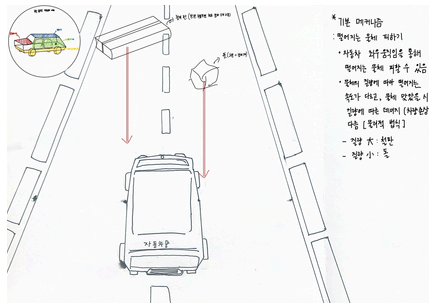
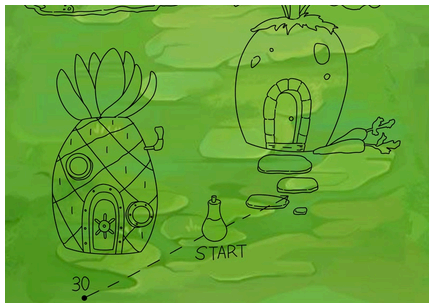
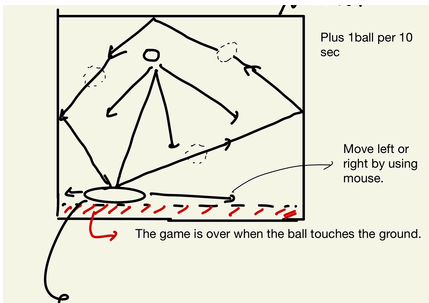
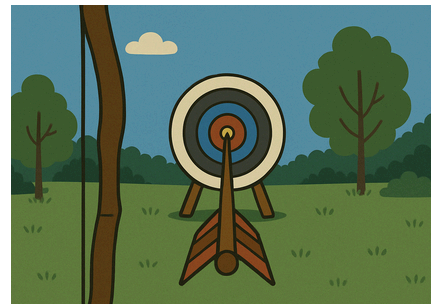
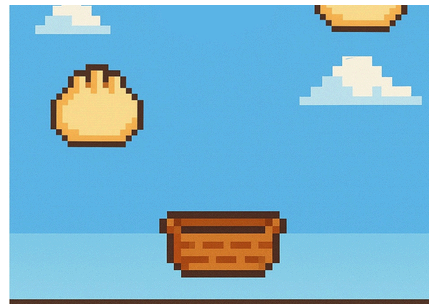
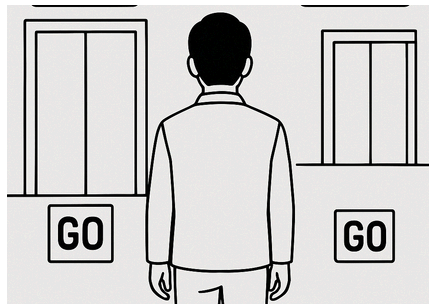
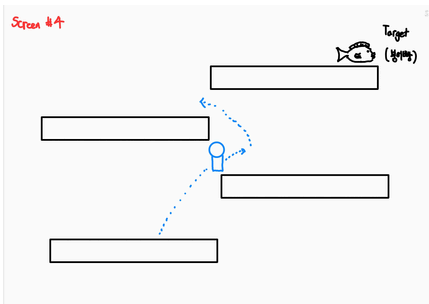


Flick football



Challenge! A+

Engineering Mathematics I: Spring 2025



Game Day (Spring 2025)

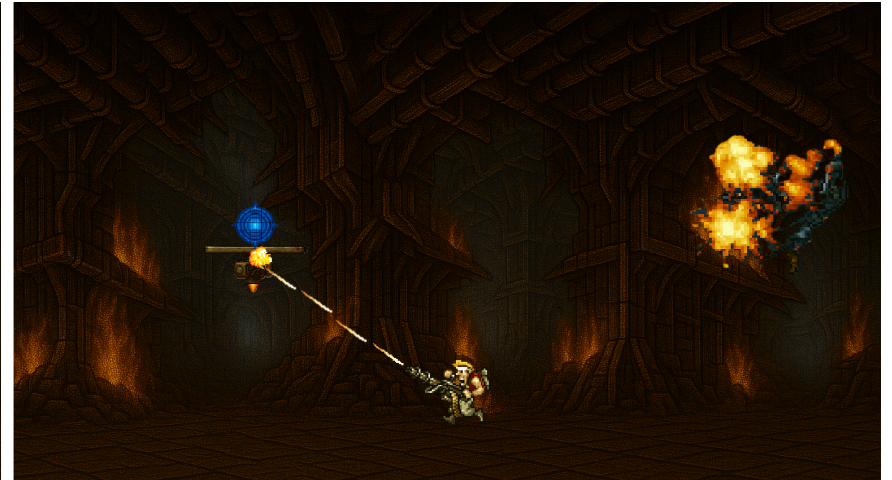


[Play video · 2:05](#)

Final Thoughts

Math Education Needs a New Medium

- AI can assist **visual design** and **coding**,
- but leaves the **story-telling** to the makers.



Creating is the life's ultimate goal.

- Learning is becoming your own **teacher**.
- Teaching is **studying** your students.
- "What I cannot create I do not understand." — **Richard Feynman**
- "The essence of mathematics is not to make simple things complicated, but to make complicated things simple." — **Stanley P. Gudder**

Thank you

Math Projects: hyosangkang.github.io